

The Tube

Topological consequences of heterotrophy in the mortal scientist framework

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Abstract

A heterotroph eats in order to reach the token supply held inside another agent. Time spent *exclusively* on extracting that supply is time not spent foraging, mating, or forming and testing hypotheses, and so carries an opportunity cost that selection can be expected to attack. The attack available is concurrency: split the harvest into a fast acquisition phase and a slow internal phase, and let the second run in parallel with everything else the agent does. The internal phase leaves a residue — the part that was not energy — and the residue must leave. Material in, refuse out. This note argues that the heterotroph is therefore, topologically, a *tube*, that the autotroph has no corresponding pressure, and that this is exactly the arrangement observed in the physical world: animals are tubes, plants are not. The argument is set inside the mortal scientist framework, where the pieces it requires — trophic access profiles, interposed media, porosity and partial namespace closure — are already present. Nothing new is assumed. The claim is that the tube is derivable, and in particular that it is a second corollary of the same proposition that already yields “intelligence is a heterotroph’s expense”. The note is a research note in the strict sense: it records an argument, a list of statements to attempt, and the empirical checks each would face. The September 11 revision adds two things. The first is a refinement of the criterion: the discriminating variable is not the rate of exchange but an occupancy ratio together with a separability, which is what distinguishes eating from breathing and explains why respiratory systems themselves are sometimes tubes. The second is a set of counterexamples proposed by M. Stay — bag-gutted animals, and heterotrophs outside the animals — which are worked through one by one. One of them is factually mistaken, four are predicted by the refined criterion, and two force a narrowing of scope from heterotrophy to phagotrophy that the argument should have carried from the start.

1 The observation

The reason for eating is to gain access to the token supply held inside the agent being consumed.

There is a distinct evolutionary advantage to minimising the time spent *exclusively* on energy extraction. If some portion of the extraction can be carried on internally to the heterotroph — so that the heterotroph can meanwhile mate, seek further food, form hypotheses about its environment, make predictions — then extraction runs concurrently with everything else the agent does, and that concurrency is worth paying for.

Hence eating has two phases:

1. the gross phase: getting the relevant material across the boundary and inside;

2. the internal phase: the remaining processes working on that material while the agent is engaged elsewhere.

The residue of the internal phase is the part that was not energy. It must leave. Material in, refuse out: this makes the heterotroph, topologically, a tube.

The autotroph has no such need. Nothing about harvesting an external source calls for a tube structure. And this is the arrangement found in the physical world.

2 What the framework already supplies

Every piece the argument needs is on the table. The claim of this note is not that a new mechanism is required but that the tube *follows*.

Access profiles. The scientist note's trophic section distinguishes a source — a persistent emitter, rate-limited, non-exclusive, non-adaptive — from prey — a linear send, exhaustible, exclusive, adaptive. Same sort, same rewrite rule, different multiplicity. Trophic type is a fact about the cut, not a new kind of thing.

The asymmetry that generates the refuse. A source emits *tokens*. Prey is a *computation*: a structured term that happens to hold a stack. Harvesting a source yields token and nothing else; breaking prey open yields token entangled with code.

Observation 1 (Refuse is not an extra postulate). The residue exists precisely because prey are computations and sources are not. It is the non-token part of a structured prey term. No assumption beyond Table 1 of the scientist note is required to produce it.

Media and interposed channels. The engine note's error model generalises `for(y <- x) P | x!(Q)` to `for(y <- x1)P | C(x1,x2)| x2!(Q)`, and ranks media by behaviour: relay, lossy, corrupting, reading, converting, with the engine at the top rung. A d -hop pathway is already constructed there.

Porosity and individuation. An individual is Chn_{in} closed, $\text{Chn}_{\text{boundary}}$ open. Boundary channels require redex pathways to internal channels or they are stranding sites. Integration depth d costs reliability p^d ; ownership is what buys depth.

The tower. The outside of one level is the inside of the next.

Namespace separation as the guarantor of non-interference. In the game note, threat offers had to live on their own names: a line signalling on a square's move channel would rendezvous with that square's receipt and play the move. Perception must not be able to act.

3 The central identification

Observation 2 (The gut is an internalised engine). The digestive apparatus is a converting medium held inside the individual's boundary namespace: a chain of interposed channels running from an intake channel to an expulsion channel, with the whole chain owned.

The mycorrhizal network was the external instance of the engine profile. The gut is the same profile drawn on the other side of the ownership line. This is not an analogy between two systems; it is one access profile appearing at two positions relative to a boundary.

The observation that makes the topology non-metaphorical is the next one.

Observation 3 (The lumen is outside). The interior of the tube is not inside the individual. It is boundary namespace: owned, but open.

A tube is therefore the construction that surrounds a region of the outside with the inside. That is the tower's generating step, applied for a specific purpose: to acquire boundary surface without acquiring exposure. The biological commonplace that the gut lumen is topologically external is, on this reading, the same fact as the framework's fractality — not a coincidence to be noted but the mechanism itself.

4 Statements to attempt

Numbering is provisional and every item below is a candidate rather than a result.

Proposition 4 (Separation is not itself yielding). *In the harvest of a structured prey term, the rewrite steps that separate token from residue are metered but token-neutral. They cost and do not pay.*

Proposition 5 (The serialisation penalty). *If separation is sequential with the scientist's control loop, the scientist pays its burn rate over the whole separation with no inquiry, no foraging and no mating during it.*

The target here is to write the foraging inequality with a sequential fraction σ and derive the Amdahl-shaped bound on sustainable harvest rate, so that the advantage of concurrency emerges as a strict inequality with an explicit threshold rather than as a preference.

Proposition 6 (Concurrency forces internalisation). *Concurrency as such is trivially available in ρ : it is parallel composition. The content is therefore not that extraction can be concurrent but where the extraction channels live. If separation runs on boundary channels, other agents may rendezvous with partially processed material. Exclusivity of a partially extracted harvest requires the separation chain to be owned.*

Proposition 6 is the ownership-buys-depth corollary specialised to digestion, and it is the step that converts a remark about scheduling into a claim about topology.

Proposition 7 (Two ends, not one). *Intake and expulsion must sit on distinct names. On a single shared channel an offer of refuse and a receipt for intake rendezvous: the organism re-ingests its residue, or the residue blocks the intake. A single-opening organism is therefore not impossible but rate-limited by a mutual exclusion — intake and expulsion must alternate on the shared name.*

This is the perception-must-not-act argument of the game note, one level up. It has an immediate empirical check, and the check is favourable: cnidarians and flatworms possess a gastrovascular cavity with a single opening and feed in discrete alternating bouts, while bilaterians with a through-gut feed continuously. The framework should predict the throughput gap from the exclusion alone.

Proposition 8 (Autotrophs are not tubes). *For source access, both of the tube’s ends lose their reason. The profile is non-exclusive, so nothing is gained by fast acquisition; the source is unstructured, so there is no residue to expel.*

The prediction is not merely that a plant needs no tube but that it should instead buy boundary area directly. A leaf and a root system are that purchase.

Corollary 9 (The second heterotroph’s expense). *The scientist note derives “intelligence is a heterotroph’s expense” from the limit-of-inquiry proposition together with structural recursion. The tube should be a second corollary of the same proposition: exclusivity together with structure forces both purchases.*

If Corollary 9 goes through it is the strongest result available here, because it makes a gross anatomical fact a consequence of an access profile. Intelligence and the tube become siblings rather than neighbours.

Proposition 10 (Transient versus persistent tubes). *A food vacuole is a tube built and dissolved per meal; a gut is one maintained. Persistence carries a maintenance cost against the metabolic stack, so there is a threshold harvest rate above which the permanent tube beats the transient one.*

Amoeba below the threshold, annelid above it. The composite-resolution proposition already says the overlap region between strategies is populated by composites rather than atoms, so a graded answer here is expected rather than a disappointment.

Proposition 11 (Gut depth tracks incommensurability). *Tokens held in a foreign colour are not nutritive as such; they must be converted. The number of interposed converting stages should scale with the colour distance between prey namespace and consumer namespace.*

Carnivores eat close relatives and have short guts. Herbivores eat far ones and have long guts, often renting whole foreign namespaces — symbionts — inside the boundary to perform conversions they cannot perform themselves. A rented engine inside the tube is a market inside an individual, and it survives the Coase objection recorded in the engine note precisely because, by Observation 3, the lumen is not the inside.

5 Rate of exchange: why breathing is not eating

Dictated 11 September 2026, as a refinement of Section 1.

Respiration is also an exchange across a boundary. Something is admitted, a fraction of it is extracted, and the remains are expelled. Yet the mammalian lung is a bag and not a tube. The dictated intuition is that the discriminating variable is the *rate* of exchange: digestive processes run orders of magnitude slower than the respiratory cycle, and it is the slowness that forces the second opening.

The intuition is pointing at something real. Rate as such is nevertheless the wrong variable, and saying why turns out to sharpen the note rather than weaken it.

5.1 Respiration goes tube exactly where one would not expect it

Ventilation is unidirectional in fish — water enters at the mouth and leaves at the opercular openings — and unidirectional in birds, where air moves the same way through the parabronchi during both phases of the cycle. Unidirectional pulmonary flow is not confined to birds: it has been demonstrated in alligators [3], in the savannah monitor [11], and in iguanas, and Farmer’s review argues from that distribution that the trait is ancestral for diapsids and therefore *not* an adaptation for high rates of gas exchange, since the animals that have it include ectotherms that do not fly; candidate functions offered instead include reducing the work of breathing, evaporative water loss and heat loss [2].

So a respiratory system can be a tube while running fast, and can be a bag while running fast. High rate is neither necessary nor sufficient. What varies across those cases is the ratio of demand to what a tidal exchanger can supply — water holds a small fraction of air’s oxygen and is far more viscous — rather than the rate itself.

5.2 Two parameters, not one

Definition 12 (Occupancy ratio). $\Lambda = \tau_{\text{proc}}/\tau_{\text{enc}}$, where τ_{proc} is the residence time of an admitted load inside the exchanger and τ_{enc} is the interval at which a further load could profitably be admitted.

Definition 13 (Separability). $\mu \in [0, 1]$ measures the degree to which residue can leave past incoming material at the same aperture without bulk transport: $\mu = 1$ for a residue miscible with the intake and free to leave by diffusion, $\mu = 0$ for a residue that must be conveyed as a bolus.

Λ is what sets the sequential fraction σ of Proposition 5: if the control loop is blocked for the whole residence, $\sigma = \Lambda/(1 + \Lambda)$, and the Amdahl-shaped bound bites only as Λ approaches and passes unity. This is the sense in which the dictated intuition is right. Slowness matters, but only relative to the opportunity structure. A process that takes a day is cheap to serialise if the next load cannot arrive for a week.

5.3 Breathing fails on separability, not on rate

The respiratory residue fraction is close to one: almost all of the mass admitted is expelled again. Bag topology survives that because $\mu \approx 1$. The mammalian lung in fact tolerates gross re-mixing — an inspired bolus on the order of a quarter of a litre mixes with something like a litre and a half of residual gas, six parts spent medium to one part fresh, every cycle [2] — and it works, because the residue is the same phase as the resource and separation happens by diffusion at the membrane rather than by transport through the aperture.

Digestion has $\mu \approx 0$. A bolus of partly extracted material cannot diffuse past incoming food, and the cost of re-mixing is not a few percent of exchange efficiency but contamination of the batch.

Proposition 14 (The tube criterion). *A tube is forced when the agent is phagotrophic — admitting bulk material with a non-absorbable fraction — and $\Lambda \gtrsim 1$ and $\mu \approx 0$. Failure of any one of the three is sufficient for a bag.*

Proposition 14 subsumes Proposition 7 rather than replacing it: the mutual exclusion on a shared name is what the agent pays, and Λ is how much of the cycle that exclusion consumes.

Corollary 15 (Amendment to Proposition 8). *Proposition 8 was stated for autotrophs and should have been stated for non-phagotrophs. Source access is one way to have nothing to expel; it is not the only way.*

6 Counterexamples, and where they land

M. Stay (personal communication, 11 September 2026) proposed a list of animals with bag-like rather than tube-like digestive systems — cnidarians, ctenophores, flatworms, xenacoelomorphs and ophiuroids — together with two cases from outside the animals: fungi, which are heterotrophs, and plants such as *Monotropa uniflora*, which parasitise fungi rather than photosynthesise. The list is the right test set, and it separates cleanly into three kinds of case: one that is factually mistaken, four that the occupancy ratio predicts, and two that force a correction of scope.

Case	Verdict	Why
Ctenophora	fails	has a through-gut
Cnidaria	predicted	low Λ ; duty-cycles when Λ rises
Platyhelminthes	predicted	buys area; gutless where osmotrophic
Xenacoelomorpha	predicted	no lumen at all; diffusion-scale bodies
Ophiuroidea	predicted	secondary loss, microphagous diets
Fungi	scope	osmotroph, not phagotroph
Mycoheterotrophs	scope	osmotroph, not phagotroph

6.1 Ctenophora: the counterexample that fails

Comb jellies are not bags. Presnell and colleagues showed by time-lapse imaging that the ctenophore gut is unidirectional and functionally tripartite, with waste expelled through terminal anal pores that are specialised to control outflow, resolving a long-standing misreading of those pores that had supported the blind-gut picture [10, 4].

This is more than the removal of a counterexample. Ctenophores branch very deep in the metazoan tree, so a through-gut there is evidence that the construction is not an inheritance of the bilaterian body plan but something arrived at from an access profile — which is the note’s thesis.

6.2 Cnidaria: the predicted duty cycle

Mean digestion time in *Aurelia aurita* has been measured at about one hour [7]. Against the encounter interval of a drifting tentacle feeder on dilute plankton this is a low Λ , and Proposition 14 accordingly predicts a bag.

The prediction with teeth is what should happen when Λ rises, and it has been observed. In *Pelagia noctiluca* ephyrae feeding on a dense and pulsed prey field, the gut saturates in about fifteen minutes while digestion takes about eighteen hours, and the authors read the mismatch as implying a diel feeding periodicity [5]. Fast fill, slow clear, forced alternation: that is Proposition 7 observed in a single organism, with the ratio measured.

6.3 Platyhelminthes and Xenacoelomorpha: area instead of length

Acoels have no gut lumen at all. The mouth opens into a syncytial digestive parenchyma without epithelial lining, in which particles are handled by phagocytosis. At body scales where the diffusion distance is a few cell diameters there is no bulk transport to arrange and μ has no work to do. The branched blind gut of a triclad is the same move at larger scale: it buys boundary area rather than a second opening, which is the purchase Proposition 8 assigns to autotrophs.

Cestodes are the sharper case. Tapeworms have no digestive system whatever and take up small organic monomers across the tegument. A cestode is an animal, a bilaterian, a heterotroph, and descended from ancestors with a through-gut — and it is not a tube, because something else did the digestion. No formulation in terms of heterotrophy can accommodate that; Proposition 14 accommodates it by the phagotrophy clause.

6.4 Ophiuroidea: secondary loss under a diet shift

Brittle stars lack an anus, as do paxillosid asteroids, and the review literature catalogues repeated independent losses of the anal opening across Bilateria, of which these are two [6, 8]. The direction of change matters: this is loss from a through-gut ancestor rather than a lineage that never built one, and it is concentrated in microphagous habits, suspension and deposit feeding on small particles, where the indigestible fraction per load is small and Λ correspondingly low.

That yields a within-clade test. Macrophagous ophiuroids — scavengers and predators taking large items — should show markedly harder duty-cycling than their microphagous congeners, and if they do not, Proposition 14 is in trouble on ground of its own choosing.

6.5 Fungi and mycoheterotrophs: the scope was stated too widely

These are not counterexamples. They are a correction, and a useful one, because they identify the clause the note had left implicit.

Heterotrophy is a claim about carbon source. The tube argument is a claim about *phagotrophy*: about admitting bulk material that contains something which will not be absorbed. A fungus secretes its enzymes outward and takes up monomers; the residue never crosses the boundary at all. Mycoheterotrophic plants do the same at one further remove, drawing fixed carbon through a mycorrhizal interface [9, 1], and *Monotropa uniflora* is the resolute case, taking everything it has through associations with Russulaceae.

Observation 16 (Osmotrophs evert the lumen). Observation 3 said the lumen is boundary namespace, owned but open. An osmotroph runs the same converting chain without the ownership: the substrate *is* the lumen, and the engine sits at the boundary rather than folded inside it. The fungus is the tube construction turned inside out — which is why the mycorrhizal network appeared in the engine note as the external instance of the very profile Observation 2 draws internally.

The cost structure is then legible rather than anomalous. Internalising the chain buys exclusivity over a partially extracted harvest, which is Proposition 6; externalising it surrenders exclusivity and in exchange never has to transport residue. An osmotroph is an agent that has taken the other side of that trade, and it should be predicted to be vulnerable exactly where the trade is worst, namely to theft of extracellular product by neighbours.

6.6 A prior statement of Proposition 5

Honesty about priority: the functional claim is already in the literature. The same review that catalogues the losses states in passing that a one-way gut processes food more efficiently and permits uptake while the animal is still digesting [6]. The contribution claimed here is therefore not the observation but the derivation — that the two openings follow from concurrency together with ownership, and that the threshold is a computable function of Λ and μ rather than a qualitative preference.

7 What to write down formally

The two-phase harvest as a term. An exclusive acquisition rendezvous on a boundary channel, in parallel with a persistent digestion chain on internal channels, terminating in a send on the expulsion channel. The existing `Forage` of the scientist note’s worked ecosystem and the `Chain` of `engine.rho` supply most of it.

The tube as a namespace shape. $\text{Chn}_{\text{boundary}}$ factors as $\text{Chn}_{\text{in}} \sqcup \text{Chn}_{\text{out}}$ with a directed redex pathway from in to out and no return path. Then *tube* is a formula in the logic rather than a designation, which is the same move already made for “being a scientist is a depth- ≥ 2 dialogue loop through a both-type channel”.

Whether the absence of a return path is forced. Coprophagy, caecotrophy and rumination are return paths, and they occur exactly where one pass leaves too much unextracted. This is cheap to state as a condition on residual yield and should be stated, since a proposition that forbids observed behaviour is worse than one that predicts it.

The occupancy ratio as a term-level quantity. Λ of Definition 12 should be recoverable from the harvest term itself: τ_{proc} as the metered cost of the digestion chain, τ_{enc} as the expected waiting time on the intake channel under the ambient offer rate. Then Proposition 14 becomes an inequality between two quantities the framework already meters, and the tube threshold is derived rather than posited.

Separability as a namespace condition. μ is not a physical parameter but a statement about whether residue and intake can share a name without rendezvous — that is, whether the sorts involved make the crossed redex impossible. Gas exchange has $\mu = 1$ because the spent medium and the resource are the same sort and the extraction happens at the wall rather than at the aperture. This wants stating as a condition on the sort discipline, at which point Proposition 14 is entirely internal.

The physical topology as a shadow. A through-gut makes the body a genus-one surface. That is real, but it is not the content. The content is the channel topology; the surface is what an embodiment of that topology in three-space looks like. The note should say so explicitly, so that it does not appear to be arguing from anatomy to mathematics.

8 Open questions

Question 1. Is the tube forced or merely favoured? Composite resolution predicts mixotrophs, and predicts that their tubes should be facultative. Is that borne out?

Question 2. Does the surface-area argument connect to the critical radius r^* of the compression-versus-autonomy result? A tube buys boundary area at fixed volume; r^* is a critical radius. There is plausibly one calculation here rather than two.

Question 3. Where does circulation enter? Beyond a certain body size the tube's yield must be distributed, which is a second internal medium with a different profile — relay rather than converting — and a different topology, branching and returning. If the tube follows from the access profile, does the vascular tree follow from the tube together with the allometric constraint?

Question 4. What is the assay analogue? Tasting is an assay run at the intake channel deciding whether to admit; vomiting is a refutation acted upon after admission. The trichotomy and budget-relative refutation should both apply, and the lossy-medium correction owed upstream to the scientist note's assay section may turn out to be the same correction.

Question 5. Does the tube give a new grading dimension, or is it already covered by the vector over typed namespaces? Suspicion: the tube is a fact about k_{chn} , which would be an independent argument for the fifth namespace.

Question 6. Does the osmotroph/phagotroph split have a framework-native characterisation? Observation 16 says the osmotroph runs the converting chain outside the ownership boundary. If that is right, the split should fall out of where the chain sits relative to $\text{Chn}_{\text{boundary}}$, and osmotrophy should carry a predictable exposure to theft that phagotrophy does not.

Question 7. Is there a second critical point at high Λ ? Ruminants and hindgut fermenters run τ_{proc} far above τ_{enc} and answer with buffering — a sequence of chambers, and in the ruminant a deliberate return path. Buffering is the standard answer to a throughput mismatch, and the question is whether the framework predicts the chamber count.

9 Filing

The natural home is the trophic-structure section of the scientist note, since the material is short and is a corollary of what is there. If Propositions 5, 7 and 11 develop enough measurable content — a threshold, a throughput ratio, a scaling law — the note stands on its own in the rho-life series instead.

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